

AYUSH SHARMA

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Summary

Computer Science Engineering (AI) graduate with hands-on experience building AI-powered applications, backend systems, and machine learning solutions. Skilled in Python, SQL, FastAPI, TensorFlow, and computer vision, with practical experience developing production-style projects involving LLM integration, REST APIs, authentication, and database design. Passionate about solving real-world problems through scalable software engineering, data-driven development, and continuous learning.

Education

Dr. A.P.J. Abdul Kalam Technical University (AKTU)

June 2022 – July 2026

Bachelor of Technology in Computer Science Engineering & Artificial Intelligence

Relevant Coursework: Artificial Intelligence, Machine Learning, Data Structures, Database Management Systems, Computer Vision

Technical Skills

- **Programming Languages:** Python, C, C++, SQL
- **Libraries & Frameworks:** NumPy, Pandas, Matplotlib, Scikit-learn, FastAPI
- **Machine Learning:** Supervised Learning, Model Training & Evaluation, Data Preprocessing
- **Deep Learning:** Neural Networks, CNNs, Model Training, TensorFlow
- **Web & Backend:** REST APIs, WebSockets, JWT/OAuth Authentication, MySQL, Firebase
- **Tools & Technologies:** Git, GitHub, Docker, Jupyter Notebook, VS Code, AI-assisted Development (Claude)
- **Soft Skills:** Analytical Thinking, Communication, Team Collaboration, Quick Learner

Projects

TokenBridge – Token-Efficient AI Proxy Middleware

Personal Project

Python, FastAPI, MySQL, JWT/OAuth, Firebase, JavaScript

- Designed and developed a multi-provider LLM routing platform (Claude, OpenAI, Gemini) with a Bring-Your-Own-Key (BYOK) model, enabling users to route AI requests through a single unified backend.
- Implemented context compression and prompt optimization techniques to reduce token usage and API costs across providers.
- Built secure authentication using JWT, OAuth, and Firebase, along with a responsive, styled frontend for chat-based interaction.

MistRoom – Privacy-First Anonymous Chat Application

Personal Project

Node.js, WebSockets, End-to-End Encryption, MySQL, Docker

- Built a privacy-first anonymous chat application with real-time messaging over WebSockets, using end-to-end encryption (E2EE) so messages stay unreadable to the server, and a zero-logging design so no message data is ever stored.
- Designed an offline peer-to-peer mesh networking mode, allowing devices to exchange messages directly with each other when no central server is reachable, improving resilience and privacy.
- Architected the backend with a MySQL-based data layer and containerized the full application using Docker Compose, applying human-led system design and AI-assisted (Claude) implementation to structure the codebase for reliable, repeatable deployment.

FaceTrackAI – Real-Time Face Tracking & Recognition System

Academic Project

Python, OpenCV, Flask, Machine Learning, Computer Vision

- Built a real-time face recognition system to automate attendance tracking with high accuracy using OpenCV and the face recognition library.
- Designed a Flask-based web application for face registration, training, and automated attendance logging, optimized for real-time video processing across multiple users.

Experience

Database Management Intern – Merlin Creations Pvt. Ltd.

July 2025 – Aug 2025

Technologies: MySQL, SQL

- Designed, developed, and maintained MySQL databases, ensuring efficient data storage and retrieval.
- Wrote and optimized complex SQL queries, improving database performance and reducing query execution time.
- Performed data import/export, backup, and recovery operations, ensuring data availability, reliability, and performance tuning.

Certifications & Achievements

- IBM SkillsBuild in Advanced IT Skills – Certificate Course on Artificial Intelligence, Grade A+ (ICT Academy, Feb 2026)
- GenAI Powered Data Analytics Job Simulation – Tata (Forage), hands-on experience in AI-driven analytics and business problem-solving
- Cybersecurity Analyst Job Simulation – Tata (Forage), focused on threat analysis, IAM fundamentals, and security risk assessment
- Data Analytics Program Certification – EXL & American India Foundation, developing skills in data analysis, interpretation, and business insights
- Active learner constantly exploring new AI/ML technologies and implementing personal projects with a strong foundation in mathematics and statistics